

S Karthik

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📍 Chittoor, India

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Profile

Integrated M.Tech Software Engineering student experienced in building backend systems using Java and Spring Boot and developing machine learning pipelines for medical image analysis. Published IEEE research on Quantum Convolutional Neural Networks (QCNN) for brain tumor detection. Interested in backend engineering, scalable systems, and applied AI solutions.

Education

VIT-AP University Sept 2022 – June 2027

Integrated M.Tech in Software Engineering

CGPA: 8.58/10.0

Sri Vivekananda Jr College June 2020 – Aug 2022

Board of Intermediate Education (MPC)

Percentage: 71.3%

Z P High School June 2019 – Mar 2020

Board of Secondary Education

Percentage: 96.3%

Publications

Brain Tumor Detection using Quantum Convolutional Neural Networks (QCNN)

S. Karthik et al., *Proceedings of IEEE OCIT 2025*.

DOI: 10.1109/OCIT66168.2025.11400476

Technical Skills & Tools

Programming: Java, Python, SQL, JavaScript

Software Development: Object-Oriented Programming, REST APIs, MVC Architecture, Modular Design

Core Computer Science: Data Structures & Algorithms, Operating Systems

Databases: MySQL, NoSQL

Machine Learning: CNN, ResNet, QCNN

Frameworks & Libraries: TensorFlow, PyTorch, PennyLane

Tools: Git, GitHub, Postman

Projects

Task Management REST API (Java, Spring Boot, MySQL) GitHub | Live

- RESTful backend service for task management built using Java, Spring Boot, and Spring Data JPA.
- Modular Controller–Service–Repository architecture implemented to support scalable task CRUD operations.
- Backend application deployed on Render with publicly accessible production API endpoints.

Brain Tumor Detection using QCNN (Python, TensorFlow, PennyLane) GitHub

- Modular machine learning pipeline created for MRI preprocessing, training, and evaluation.
- Hybrid EfficientNet-B0 + QCNN architecture implemented using Python and PennyLane.
- Model achieved **90% classification accuracy**; research published in IEEE OCIT 2025.

Brain Tumor Classification using Hybrid QCNN with ResNet GitHub

- Hybrid deep learning pipeline combining pretrained ResNet feature extraction with QCNN classification.
- Training performed on **7,000+ MRI images** spanning four tumor categories.
- Validation accuracy of **91%** achieved using transfer learning and regularization techniques.

Certifications

Oracle Cloud Infrastructure 2025 Certified Foundations Associate – Oracle University

Fundamentals of Digital Marketing – Google Digital Garage

HTML & CSS for Web Development – Certification

Introduction to Machine Learning with R – Simplilearn

Introduction to Statistics – Stanford University (Coursera)

Languages

English — Tamil — Telugu — Hindi